

Qwen3.5-9B for Agentic Orchestration & Coding

Where a 9B model fits, where it breaks, and how Qwen3.6-27B and Qwen3.8-27B compare

Trainer: Lakshminarayanan

Date: Sep-Oct 2026

Three models at a glance

All three are Apache 2.0, dense, multimodal and self-hostable with 262K native context



Qwen3.5-9B

Feb 2026 · Router / worker

- 9B dense, 32 layers
- ~6–10 GB at 4-bit
- Strong single-step tool calling
- Best for routing, extraction, classification

31.3

SWE-bench Pro
(third-party run)



Qwen3.6-27B

Apr 2026 · Agentic coder

- 27B dense, 64 layers
- ~16 GB at 4-bit (one 24 GB GPU)
- Beats Qwen3.5-397B on coding benchmarks
- `preserve_thinking` across turns

53.5

SWE-bench Pro
(vendor-reported)



Qwen3.8-27B

14 Aug 2026 · Agent planner

- 27B dense, 48 DeltaNet + 16 full-attention layers
- Runs on one 24 GB GPU at 4-bit
- SWE-bench Pro 61.7, OSWorld 84.3
- Numbers still vendor-reported

61.7

SWE-bench Pro
(vendor-reported)

Where a 9B model becomes risky for agentic orchestration

Five failure zones to design around



Long-horizon planning

Multi-day plans, multi-file systems, many coordinated agents. It can lose the thread as context grows.



Complex multi-tool reasoning

Choosing among many tools, remembering outputs, retrying after failures. Reliability drops quickly.



Ambiguous user goals

Wrong tool, hallucinated details, unsafe assumptions, plausible but wrong JSON.



High-stakes, irreversible actions

Deleting data, moving money, production changes. Needs approval, validation, audit and rollback.

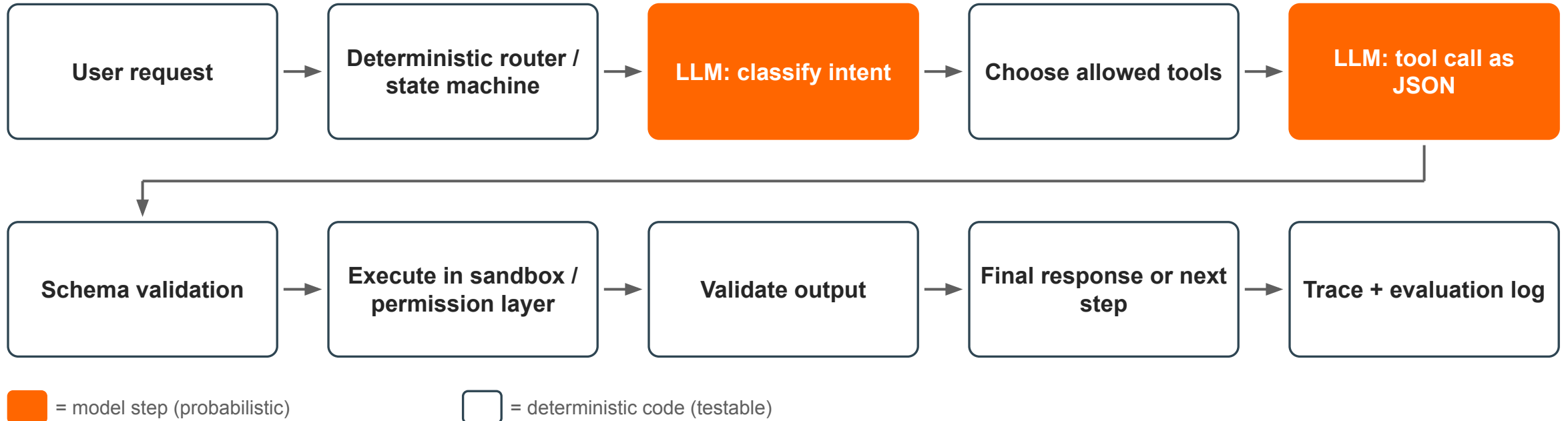


Prompt injection

Agents act on emails, documents and tool outputs. Smaller models are easier to steer with hostile text.

The safe pattern: deterministic flow, bounded LLM

Yes, a 9B can orchestrate if constrained. The workflow engine owns control flow; the model only fills specific decision points



Safer: 8 of 10 steps are plain code. The model proposes; validators and the permission layer decide.

Risky: User request → LLM freely decides and acts. No checkpoint between the model's guess and a real side effect.

Best agentic use cases for Qwen3.5-9B

Short reasoning chains and clear schemas work; open-ended autonomy does not

Excellent fit	
Customer support triage	Structured outcome
Intent classification	Short reasoning chain
Data extraction / form filling	Clear input/output schema
SQL generation	With examples + validation
API tool calling	If the tool set is small
RAG Q&A	If retrieval is good
Log summarisation / email triage	Bounded scope
Workflow step completion	One- or two-step logic

Conditional fit	
Multi-step customer workflow	Limit steps, validate each
Basic coding agent	Sandbox, tests, human review
Internal automation	Allowlisted tools only
Data pipeline assistant	Deterministic validation
Multi-agent routing	Router, not planner
Research assistant	Good retrieval + citations

Poor fit without safeguards	
Autonomous multi-file SWE	Loses state
Production infra control	Unsafe without permissions
Financial transactions	High risk
Legal / medical decisions	Not as sole decider
Open-ended multi-agent debate	Hard to control
Long autonomous research loops	Context/state issues

Seven rules for deploying a 9B model in an agent

Reliability comes from the harness, not from the model alone

1

Structured output

JSON only, validated against a schema before acting

2

Limit the toolset

3–5 tools per context, small schemas, clear names

3

Low temperature

0 to 0.3 for tool calls; no creative variance

4

Few-shot examples

2–5 worked examples improve reliability sharply

5

Separate reason from act

Propose → validate → permit → execute

6

Retries and fallbacks

Retry 1–2x, constrained decoding, escalate to a bigger model

7

Evaluate continuously

Tool-call accuracy, JSON validity, override rate, cost per task

Model proposes this, code decides

```
{
  "decision": "create_ticket",
  "confidence": 0.93,
  "tool": "create_support_ticket",
  "arguments": {
    "subject": "Login issue",
    "priority": "medium",
    "customer_email":
      "user@example.com"
  }
}
```

- ✓ schema valid?
- ✓ tool on allowlist?
- ✓ confidence ≥ threshold?
- ✓ caller has permission?

Head-to-head: specifications and deployment

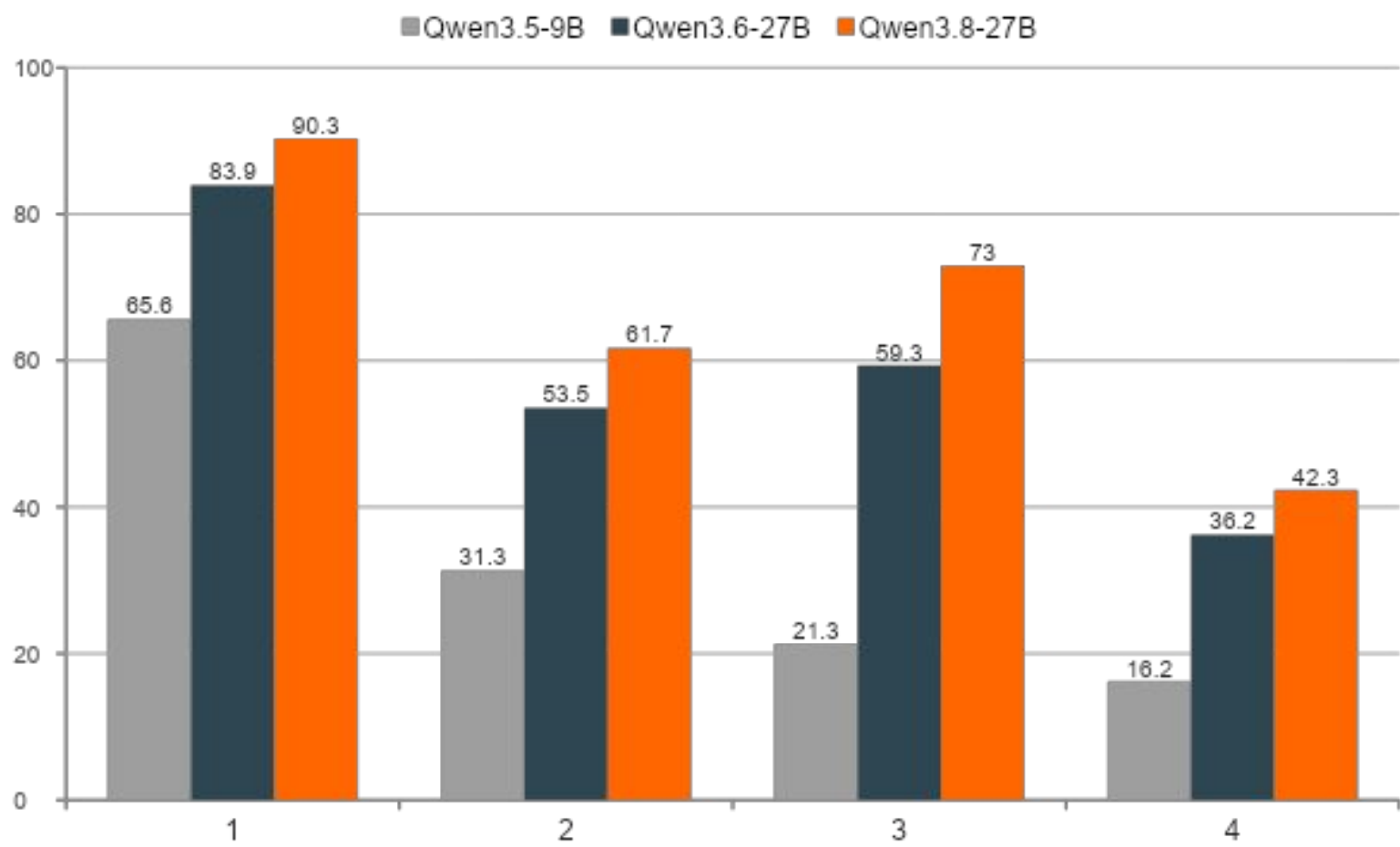
Same family and context length; the 27B models trade 3x memory for far stronger agent behaviour

Attribute	Qwen3.5-9B	Qwen3.6-27B	Qwen3.8-27B
Released	February 2026	April 2026	14 August 2026
Parameters	9B dense (~10B with vision)	27B dense (~28B with vision)	27B dense (28B with vision)
Architecture	Hybrid Gated DeltaNet + Gated Attention, 32 layers	Same hybrid, 64 layers	48 Gated DeltaNet + 16 full-attention layers
Context	262K native, ~1M with YaRN	262K native, ~1M with YaRN	262K native, ~1M with YaRN
Inputs	Text, image, video	Text, image, video	Text, image, video
Agent features	Tool calling, thinking / non-thinking	+ preserve_thinking across turns	+ strong computer / mobile use
Memory (BF16 weights)	~18–20 GB	~55 GB	~55.6 GB
Memory (4-bit)	~6–10 GB	~16 GB · one 24 GB GPU	One 24 GB GPU
License	Apache 2.0	Apache 2.0	Apache 2.0

Memory figures are weights only; add KV cache for long context, batching and concurrent agents. Sources: Qwen model cards on Hugging Face; regolo.ai Qwen3.8-27B benchmark review (Aug 2026).

Coding: the 27B models are in a different class

Agentic coding scores (% , higher is better)



+22 pts

SWE-bench Pro: 3.6-27B vs 9B (53.5 vs 31.3)

3.4x

Terminal-Bench: 3.8-27B vs 9B (73.0 vs 21.3)

77.2%

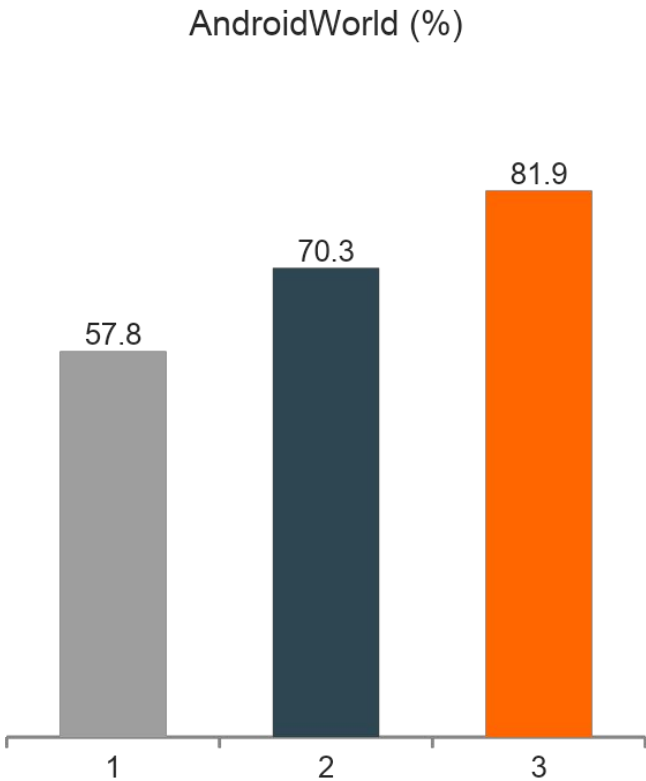
SWE-bench Verified for 3.6-27B vs 53.2% for 9B

**Terminal-Bench 2.1 for 9B and 3.8; 2.0 for 3.6. 9B SWE-bench, Terminal-Bench and NL2Repo are third-party runs (OpenHands / Terminus-2, Ornith-1.0-9B model card). Others are vendor-reported with Qwen's own scaffolds. Harnesses differ: treat gaps as directional.*

Agentic orchestration: what the benchmarks say

The 9B is good at single tool calls; the 27B models hold up over long, multi-step, multi-screen tasks

Benchmark	Why it matters for agents	3.5-9B	3.6-27B	3.8-27B
TAU2-Bench	Multi-turn tool use with a simulated user	79.1	n/r	n/r
BFCL-V4	Function-calling accuracy	66.1	n/r	n/r
Claw-Eval (avg)	Real agent tasks; Pass^3 = consistency	53.2†	72.4 (Pass^3 60.6)	n/r
IFBench	Obeys precise constraints and formats	64.5	n/r	79.5
AndroidWorld	Operates a mobile device	57.8	70.3	81.9
OSWorld-Verified	Operates a real desktop	41.8	n/r	84.3
GPQA Diamond	Hard reasoning for planning	81.7	87.8	89.2



One of few agent benchmarks reported for all three. Steady climb with each generation.

n/r = not reported on the official card. †Third-party run (Ornith-1.0-9B card). All other figures vendor-reported by Qwen; 3.8-27B not yet independently reproduced.

Six ways the 27B models are better for orchestration

Each gap maps directly to a 9B failure zone



Longer, steadier plans

Claw-Eval Pass³ 60.6 (3.6) rewards passing the same task three times; NL2Repo 36–42 vs 16 for 9B.



Survives multi-tool loops

Terminal-Bench 59–73 vs ~21: hours of tool calls, errors and retries without losing state.



Follows constraints

IFBench 79.5 (3.8) vs 64.5: fewer malformed tool calls and schema violations.



Keeps reasoning across turns

3.6 adds preserve_thinking: earlier reasoning is reused, cutting repeated thinking and tokens.



Acts on screens

OSWorld 84.3 vs 41.8 and AndroidWorld 81.9 vs 57.8: usable for RPA-style agents.



Can be the planner

Strong enough to own task decomposition, while the 9B stays the fast router beneath it.

Coding use cases: suitability heatmap

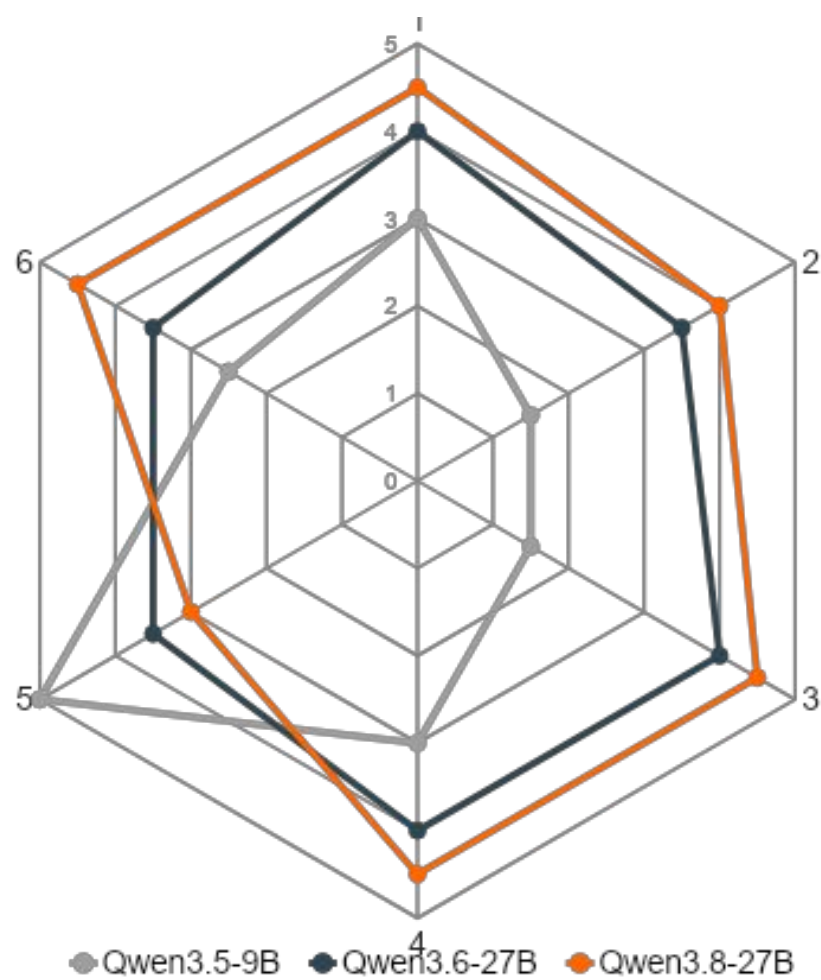
Our assessment from the published benchmarks; validate on your own repositories

Coding use case	3.5-9B	3.6-27B	3.8-27B	Evidence
Snippet / function generation	Good	Strong	Very strong	LiveCodeBench 65.6 / 83.9 / 90.3
Unit tests and code explanation	Good	Strong	Strong	Bounded scope; 9B fine with review
Single-file bug fix	Fair	Strong	Very strong	Needs reading context and running code
Multi-file repo change (SWE-style)	Weak	Strong	Very strong	SWE-bench Pro 31 / 54 / 62
Terminal / DevOps agent	Weak	Strong	Very strong	Terminal-Bench ~21 / 59 / 73
Build a repo from a spec	Weak	Good	Good	NL2Repo 16 / 36 / 42; still hard for all
Front-end from mockup or screenshot	Fair	Strong	Very strong	QwenWebBench 1487 (3.6); SWE-MM 38.6 (3.8)
Low-latency IDE autocomplete	Very strong	Good	Fair	Smaller and faster; 3.8 thinks verbosely by default

Ratings are a judgement call combining vendor and third-party benchmarks with deployment trade-offs; they are not a single published score.

Suitability profile: agentic orchestration and coding

Relative scores on a 1–5 scale



Qwen3.5-9B

Wins only on cost and latency. Keep it on narrow, validated steps.

Qwen3.6-27B

Balanced default: strong coding with proven, widely tested weights since April.

Qwen3.8-27B

Highest ceiling for planning and computer use; newest, least independently verified.

Scores are our relative assessment derived from the benchmark slides, not measured values.

What the 27B upgrade costs you, and where 9B still wins

Bigger is better for hard steps, not for every step

Cost of moving to 27B

3x

parameters and roughly 3x BF16 memory (~55 GB vs ~19 GB)

3.97 s

time to first token for Qwen3.8-27B (xhigh) on Alibaba's API

46.9 t/s

output speed on the same API, below the peer median of ~91 t/s

Also: 3.8's default reasoning can be verbose, and its benchmarks await independent validation.

Where Qwen3.5-9B still wins



Latency

Fast enough for inline routing and IDE completions



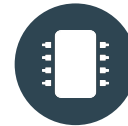
Throughput

More concurrent agents per GPU; smaller KV footprint



Cost per task

Cheapest option for high-volume classification and extraction



Edge / prosumer GPUs

Runs in ~6–10 GB at 4-bit

Latency and speed: Artificial Analysis, Qwen3.8 27B (xhigh), Aug 2026. Self-hosted speed varies widely with quantisation and speculative decoding.

Recommended: a tiered model strategy

Route every step to the smallest model that can do it safely

Deterministic workflow engine + policy engine (allowlist · schema · permissions · budget) + trace logging



Tier 1 · Qwen3.5-9B

- Classify intent, extract entities
- Pick one action from 3–5 tools
- Generate structured tool calls
- Bulk of calls, lowest cost



Tier 2 · Qwen3.6 / 3.8-27B

- Plan and decompose tasks
- Multi-file coding, terminal agents
- Resolve ambiguous requests
- Screen / computer-use agents



Tier 3 · Human or frontier

- Irreversible or regulated actions
- Money, production, legal, medical
- Repeated Tier 2 failure
- Always approved and audited

Escalate when: confidence below threshold · invalid JSON after 2 retries · more than N steps · any write to a sensitive system

Decision guide

Go / no-go for Qwen3.5-9B, and when to step up

Use Qwen3.5-9B when

- The workflow is limited and repeatable
- Tools are few and well defined
- Outputs can be validated with a JSON schema
- Actions are reversible or low-risk
- Logging, evaluation and human approval exist
- Latency and cost per call dominate

Step up to Qwen3.6-27B when

You need multi-file or terminal coding agents today, on one 24 GB GPU, with weights the community has tested since April.

Step up to Qwen3.8-27B when

You need the strongest planning, instruction following or computer-use agents, and can run your own evaluation first.

Never rely on any single model when

Mistakes are costly, actions cannot be validated before execution, or security is critical.

Thank You

Questions and discussion